Question 2:

```
import pandas as pd
import plotly.express as px

from google.colab import drive
drive.mount('/content/drive')

# Read the dataset
df = pd.read_csv("/content/drive/MyDrive/DATA/olympic_medals.csv")

# Show only the first few rows
print(df.head())

# Keep rows from year 2000 and later
df_2000 = df[df["Year"] >= 2000].copy()

# Show first few rows after filtering
print(df_2000.head())

# Parallel categories plot
fig = px.parallel_categories(
    df_2000,
    dimensions=["Gender", "Medal"],
    color="Year",
    title="Proportions of Medal Type for Each Gender Since 2000"
)

fig.show()
```

```
  Gender       Event Location  Year Medal                  Name Nationality  \
0      M  10000M Men      Rio  2016     G          Mohamed FARAH         GBR
1      M  10000M Men      Rio  2016     S  Paul Kipngetich TANUI         KEN
2      M  10000M Men      Rio  2016     B           Tamirat TOLA         ETH
3      M  10000M Men  Beijing  2008     G        Kenenisa BEKELE         ETH
4      M  10000M Men  Beijing  2008     S         Sileshi SIHINE         ETH

     Result
0  25:05.17
1  27:05.64
2  27:06.26
3  27:01.17
4  27:02.77
  Gender       Event Location  Year Medal                  Name Nationality  \
0      M  10000M Men      Rio  2016     G          Mohamed FARAH         GBR
1      M  10000M Men      Rio  2016     S  Paul Kipngetich TANUI         KEN
2      M  10000M Men      Rio  2016     B           Tamirat TOLA         ETH
3      M  10000M Men  Beijing  2008     G        Kenenisa BEKELE         ETH
4      M  10000M Men  Beijing  2008     S         Sileshi SIHINE         ETH

     Result
0  25:05.17
1  27:05.64
2  27:06.26
3  27:01.17
4  27:02.77
```

Question 4:

```
# Figure 1: lm

library(ggplot2)
library(mgcv)   # needed for gam

cars93 <- MASS::Cars93

head(cars93)

ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
  geom_point(color = "grey60") +
  geom_smooth(
    se = TRUE,
    method = "lm",
    color = "#8fe388"
  ) +
  scale_x_continuous(
    name = "price (USD)",
    breaks = c(20, 40, 60),
    labels = c("$20,000", "$40,000", "$60,000")
  ) +
  scale_y_continuous(name = "fuel-tank capacity\n(US gallons)") +
  ggtitle("Smoothing Method: lm") +
  theme(
    plot.title = element_text(size = 14, color = "#8fe388")
  )
```

```
> head(cars93)
  Manufacturer   Model    Type Min.Price Price Max.Price MPG.city MPG.highway
1        Acura Integra   Small      12.9  15.9      18.8       25          31
2        Acura  Legend Midsize      29.2  33.9      38.7       18          25
3         Audi      90 Compact      25.9  29.1      32.3       20          26
4         Audi     100 Midsize      30.8  37.7      44.6       19          26
5          BMW    535i Midsize      23.7  30.0      36.2       22          30
6        Buick Century Midsize      14.2  15.7      17.3       22          31
             AirBags DriveTrain Cylinders EngineSize Horsepower  RPM
1               None      Front         4        1.8        140 6300
2 Driver & Passenger      Front         6        3.2        200 5500
3        Driver only      Front         6        2.8        172 5500
4 Driver & Passenger      Front         6        2.8        172 5500
5        Driver only       Rear         4        3.5        208 5700
6        Driver only      Front         4        2.2        110 5200
  Rev.per.mile Man.trans.avail Fuel.tank.capacity Passengers Length Wheelbase
1         2890            Yes               13.2          5    177       102
2         2335            Yes               18.0          5    195       115
3         2280            Yes               16.9          5    180       102
4         2535            Yes               21.1          6    193       106
5         2545            Yes               21.1          4    186       109
6         2565             No               16.4          6    189       105
  Width Turn.circle Rear.seat.room Luggage.room Weight  Origin           Make
1    68          37           26.5           11   2705 non-USA Acura Integra
2    71          38           30.0           15   3560 non-USA  Acura Legend
3    67          37           28.0           14   3375 non-USA       Audi 90
4    70          37           31.0           17   3405 non-USA      Audi 100
5    69          39           27.0           13   3640 non-USA      BMW 535i
6    69          41           28.0           16   2880     USA Buick Century
```

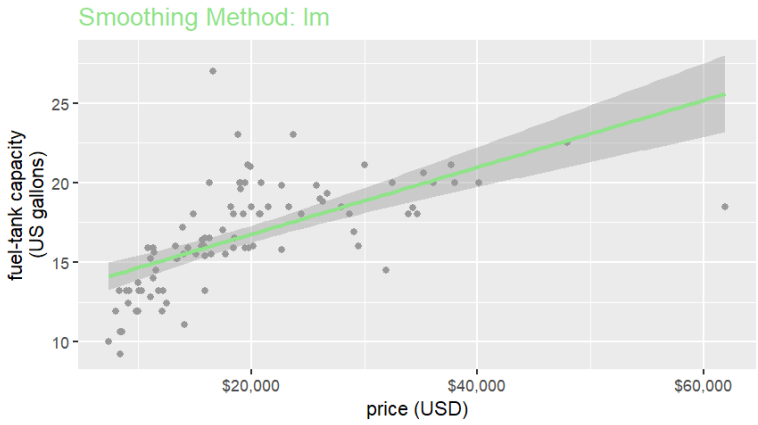


Figure 2: glm

```
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
  geom_point(color = "grey60") +
  geom_smooth(
    se = TRUE,
    method = "glm",
    color = "#fe8d6d"
  ) +
  scale_x_continuous(
    name = "price (USD)",
    breaks = c(20, 40, 60),
    labels = c("$20,000", "$40,000", "$60,000")
  ) +
  scale_y_continuous(name = "fuel-tank capacity\n(US gallons)") +
  ggtitle("Smoothing Method: glm") +
  theme(
    plot.title = element_text(size = 14, color = "#fe8d6d")
  )
```

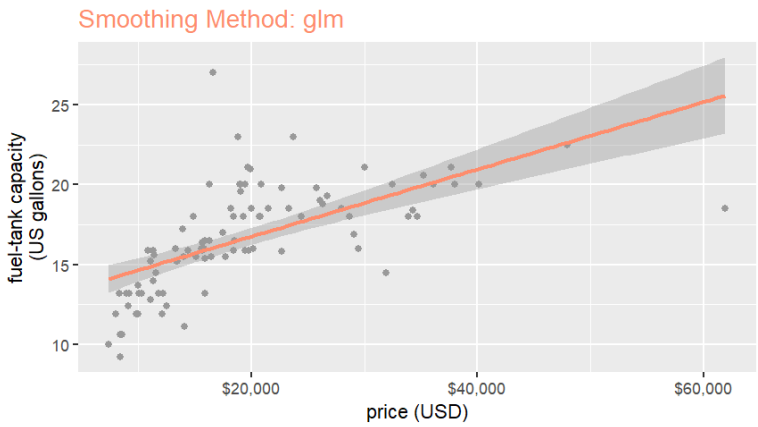
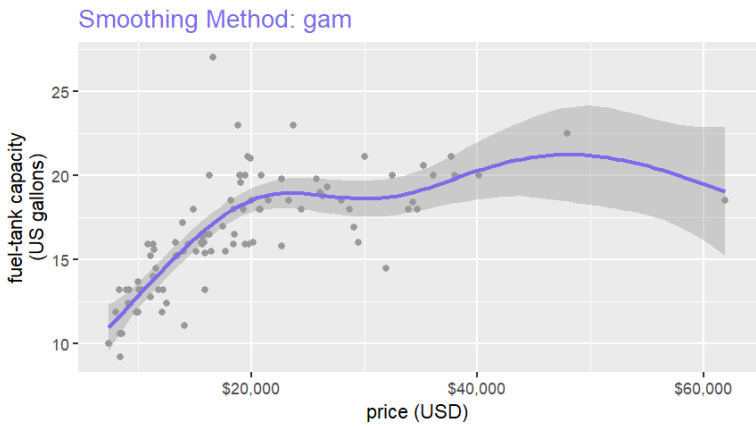


Figure 3: gam

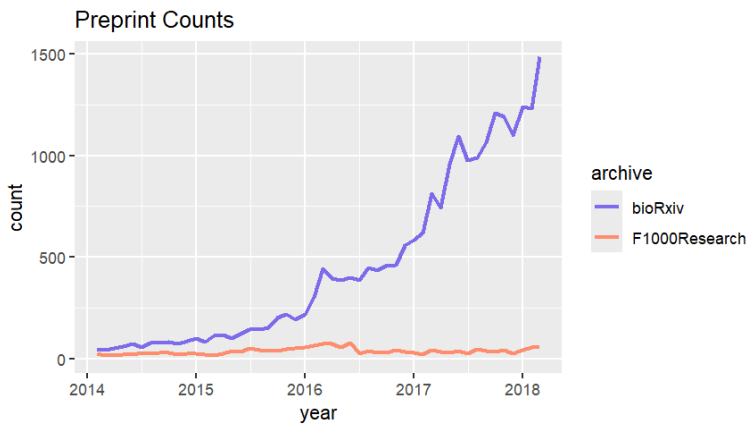
```
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
  geom_point(color = "grey60") +
  geom_smooth(
    se = TRUE,
    method = "gam",
    formula = y ~ s(x),
    color = "#7c6bea"
  ) +
  scale_x_continuous(
    name = "price (USD)",
    breaks = c(20, 40, 60),
    labels = c("$20,000", "$40,000", "$60,000")
  ) +
  scale_y_continuous(name = "fuel-tank capacity\n(US gallons)") +
  ggtitle("Smoothing Method: gam") +
  theme(
    plot.title = element_text(size = 14, color = "#7c6bea")
  )
```



Question 5:

```
> head(preprint_growth)
# A tibble: 6 × 3
  archive           date       count
  <chr>             <date>     <int>
1 arXiv q-bio       2007-01-01    40
2 Nature Precedings 2007-01-01     3
3 F1000Research     2007-01-01     0
4 PeerJ Preprints   2007-01-01     0
5 bioRxiv           2007-01-01     0
6 Winnower          2007-01-01     0
>
```

```
> head(preprint_full)
# A tibble: 6 × 3
  archive           date       count
  <chr>             <date>     <int>
1 arXiv q-bio       2007-01-01    40
2 Nature Precedings 2007-01-01     3
3 arXiv q-bio       2007-02-01    44
4 arXiv q-bio       2007-03-01    55
5 arXiv q-bio       2007-04-01    41
6 arXiv q-bio       2007-05-01    59
>
```


```
> head(preprint_full)
# A tibble: 6 × 3
  archive           date       count
  <chr>             <date>     <int>
1 arXiv q-bio       2007-01-01    40
2 Nature Precedings 2007-01-01     3
3 arXiv q-bio       2007-02-01    44
4 arXiv q-bio       2007-03-01    55
5 arXiv q-bio       2007-04-01    41
6 arXiv q-bio       2007-05-01    59
>
```